

Adam Erler

He/Him • Data Scientist

adam.erler101@gmail.com • (224) 388-8746 • linkedin.com/in/adam-erler • aerler101-1.github.io

SUMMARY

Data scientist with expertise in predictive modeling, NLP pipelines, and end-to-end ML systems, specializing in K-12 education data and policy insights. Build production data infrastructure, calibrated predictive models, and stakeholder-facing dashboards that turn unstructured procurement and governance data into market intelligence. Comfortable owning a problem from FOIA acquisition through schema design, model deployment, and presentation to executive audiences.

EXPERIENCE

Center for Education Market Dynamics (CEMD) — Remote

06/2025 – Present

Data Scientist

- Built a Dockerized NLP/LLM pipeline running on AWS EC2 (Ollama + Qwen2.5) that extracts structured procurement data from FOIA-obtained PDFs into a Pydantic schema, with OCR fallback for scanned documents, multi-vendor page segmentation, per-vendor LLM calls, retry logic, and concurrency controls.
- Developed a predictive ML model for K-12 curriculum adoption at the (publisher, series) grain using gradient boosting with district-blocked GroupShuffleSplit, additive labels, and explicit data-leakage guards; produced calibrated district-level scores and ranked outreach lists by subject and region.
- Designed and shipped an interactive market intelligence dashboard (HTML/JS) with a natural-language query interface backed by the Claude API via a Cloudflare Worker proxy, plus map, district profile, and state profile views for stakeholder exploration.
- Owned AWS infrastructure including QuickSight-to-PostgreSQL connectivity (VPC, security groups, IAM) and a GitHub Actions deployment workflow to S3 + CloudFront via OIDC, with automated cache invalidation.
- Ran NLP analyses on district board meetings, RFPs, and strategic plans using TF-IDF, LDA topic modeling, and K-means clustering to surface intermediary influence patterns (EdReports, ELSF, Rivet, NCTM) and vendor concentration signals.
- Delivered executive-facing research, including a leadership PowerPoint and one-page memo summarizing the curriculum market movement prediction project, K-5 ELA market share analyses, and SEA Assessment Signals dashboards with monthly narrative summaries.
- Built a brand-aligned PPT generation system (CEMD palette and Montserrat typography) that converts analyses into stakeholder decks with analytical, insight-driven slide headlines.

Education Strategy Consulting — Remote

06/2023 – 06/2025

Data Analyst

- Designed and managed a PostgreSQL database to store and process multi-state education data, optimizing query performance for advanced analytics and reporting.
- Developed an automated API interface for programmatic Google searches, achieving an 80% reduction in workflow time for related tasks.
- Integrated state-level education data with NCES GIS datasets to create geospatial visualizations, enabling deeper insights into regional education trends.
- Developed data visualizations to support statewide education policy decisions, enabling data-driven strategy adjustments for educational programs.
- Managed project versioning with Git, ensuring seamless collaboration and project continuity across the codebase.

Concept Schools — Chicago, IL

08/2019 – 06/2023

Teacher, Data Analyst

- Authored two grants securing over \$100,000 in funding for Project Lead The Way (PLTW) and Game Design classes.
- Implemented data-driven instructional strategies that improved student performance on state tests by at least 5%.
- Built internal data dashboards in Google Data Studio / Looker to support teacher-led data exploration.
- Developed a four-year computer science program with multiple pathways, resulting in student certifications in Unity 3D, Google, and PLTW.
- Mentored new staff daily, supporting them in meeting professional goals and integrating into the team.

Hammond Academy of Science and Technology — Hammond, IN

06/2014 – 07/2019

Grade Chair, Teacher

- Taught Algebra 1, Geometry, Web Design, and Computer Science Principles.
- Managed day-to-day operations for 9th and 10th grade teachers.
- Authored grants totaling over \$30,000 to fund a new computer science department.

OUTREACH

QuarkNet — Hammond, IN

06/2014 – 08/2023

Lead Teacher

- Orchestrated international workshops connecting high schools worldwide with live data from the Large Hadron Collider.
- Collaborated with Fermilab's Silicon Detector Laboratory to identify and address production errors in readout chips for the Forward Pixel Detector at CMS.
- Honored by the NWI Society of Innovators in 2017 for community contributions.

EDUCATION

Bellevue University — Bellevue, NE

Master of Science, Data Science

Purdue University — Hammond, IN

Master of Science, Mathematics

Southern Illinois University — Carbondale, IL

Bachelor of Science, Mathematics

SKILLS

Languages & Libraries: Python (pandas, scikit-learn, XGBoost, LightGBM, PyTorch, Pydantic), SQL, JavaScript, R, MATLAB, HTML/CSS, PySpark

ML & NLP: Predictive modeling, gradient boosting, survival analysis, calibration, permutation importance, TF-IDF, LDA, K-means, local LLM inference (Ollama/Qwen2.5), Claude API

Data Engineering & Infra: Docker, AWS (EC2, S3, CloudFront, QuickSight, IAM/OIDC), PostgreSQL, GitHub Actions, Cloudflare Workers, Apache Airflow, OCR (Tesseract), Git

Visualization & BI: Power BI, Tableau, Looker, QuickSight, Seaborn, Matplotlib, Plotly, D3, Chart.js, custom interactive HTML dashboards

Domain: K-12 curriculum adoption, assessment procurement, FOIA workflows, NCES/district demographics, education market intelligence